INCH POUND
MIL-PRF-1/22J
w/AMENDMENT 2
21 May 2024
SUPERSEDING
MIL-PRF-1/22J
w/AMENDMENT 1
18 March 2008

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING
TYPES 1B35A, 1B37A, 6162, AND 5864

Inactive for new design after
23 September 2014.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the electron tubes described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: ATR, frequency: 1B35A - - - Fo = 9,300 MHz, incident power 250 kw
1B37A - - - Fo = 8,750 MHz
6162 - - - Fo = 9,080 MHz
5864 - - - Fo = 9,375 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Du	TA	Alt
Unit:	kw	- - -	°C	ft
Maximum:	250	0.001	+100	10,000
Minimum:	4	- - -	-40	- - -

(See note 1) of table I

PHYSICAL CHARACTERISTICS: See figure 1.

TEST CONDITIONS: See note 2.

Parameter:	F	tp	prr	Du	Incident power	σ'
Unit:	MHz	μs	pps	- - -	kw	- - -
Test condition 1:	Fo ± 3%	1.0 ± 0.1	1,000	0.001	200	1.2 (max)
Test condition 2:	Fo ± 3%	1.0 ± 0.1	1,000	- - -	20	1.03

GENERAL:

Qualification - Required.

AMSC N/A
FSC 5960
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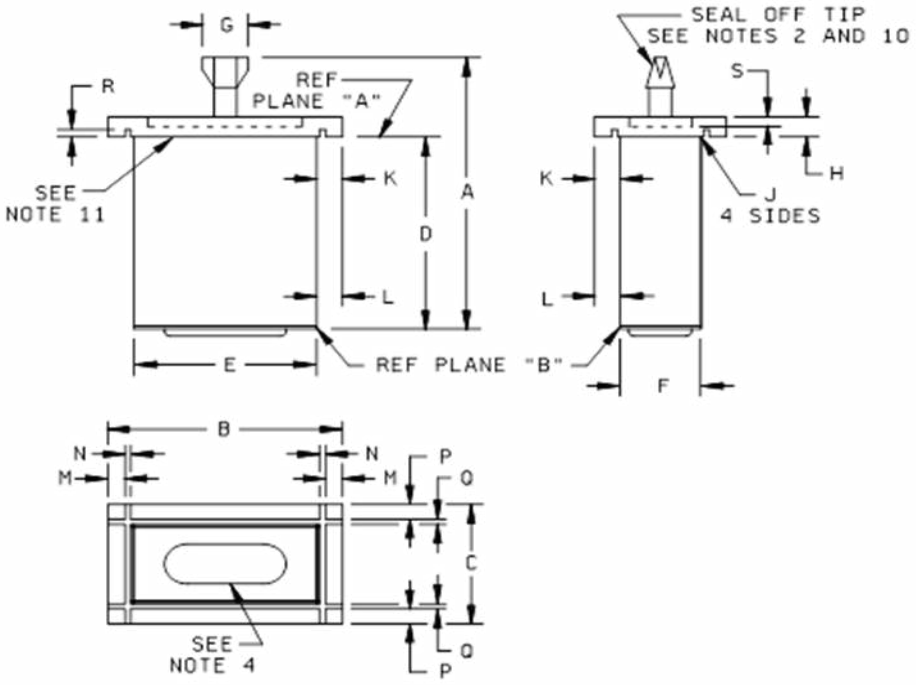
Table I. Testing and Inspection.

Inspection	MIL-STD-1311 method	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
Qualification inspection							
Degradation due to vibration	4021	-		- - -	- - -	- - -	- - -
Loaded Q (ATR tubes)	4461	-		QL	- - -	6.5	- - -
High-level VSWR	4474	-	Test condition 2	σ	- - -	1.10	- - -
Conformance inspection, part 1							
Tuning susceptance	4482	-	Fo $\pm$ 0.1% MHz	b = B/Yo	- - -	$\pm$ 0.06	- - -
Firing time	4486	3		t	- - -	10	sec
Arc loss	4488	3		La	- - -	0.8	dB
Conformance inspection, part 2							
Dielectric material strain	4101			- - -	- - -	- - -	- - -
Normalized conductance	4484	-	Fo $\pm$ 0.1% MHz	g = G/Yo	- - -	0.1	- - -
Recovery time	4471	-	Test condition 1	t	- - -	10	μs
Conformance inspection, part 3							
Life test	- - -	-	Group D; test condition 1				
Type 1B35A		-		- - -	500	- - -	Hrs
Type 1B37A		-		- - -	500	- - -	Hrs
Type 6162		-		- - -	500	- - -	Hrs
Type 5864		-		- - -	1,000	- - -	Hrs
Life-test end points:	- - -						
Arc loss	4488	3		La	- - -	1.0	dB
Normalized conductance	4484	-	Fo $\pm$ 0.1% MHz	g = G/Yo	- - -	0.1	- - -
Tuning susceptance	4482	-	Fo $\pm$ 0.1% MHz	b = B/Yo	- - -	$\pm$ 0.06	- - -
Recovery time	4471	-	Test condition 1; VSWR > 10 dB	t	- - -	20	μs
Temperature cycling (nonoperating)	1027	-		- - -	- -	- - -	- - -
Periodic-check tests							
Temperature cycling life-test end point	1027	3	Group D; 10 cycles (min)	- - -	- - -	- - -	- - -

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NOTES:

1. For 500 hours life, the duty cycle is 0.001 at pulse width of 1.0 μs. Operation at greater than 1.0 μs pulse width at 250 kw is permissible but may result in decreased life.
2. For electrical tests, the tubes shall be mounted as shown on figure 2. A suitable gasket shall be placed between the tube and the tube seat.
3. This test shall be performed at least 168 hours after pumping, and at least 24 hours after any previous discharge.



Dimensions in inches with metric equivalents (mm) in parentheses.					
Letter	Minimum	Maximum	Letter	Minimum	Maximum
Conformance inspection, part 1					
B	1.297 (32.94)	1.303 (33.10)	K	0.142 (3.61)	---
C	0.797 (20.24)	0.803 (20.40)	L	0.138 (3.53)	---
D	1.296 (32.92)	1.302 (33.07)	S	---	0.078 (1.98)
J	---	0.020 (0.51)			
Dim M through R for 1B37A only					
M	0.103 (2.62)	0.111 (2.82)	Q	0.030 (.76)	0.040 (1.02)
N	0.030 (0.76)	0.040 (1.02)	R	0.035 (.89)	0.045 (1.14)
P	0.103 (2.62)	0.111 (2.82)			
Conformance inspection, part 2					
A	---	1.813 (46.05)	G	---	0.250 (6.35)
E	0.990 (25.15)	1.010 (25.65)	H	0.117 (2.97)	0.133 (3.38)
F	0.490 (12.45)	0.510 (12.95)			

FIGURE 1. Outline drawing of electron tube types 1B35A, 1B37A, 6162 and 5864

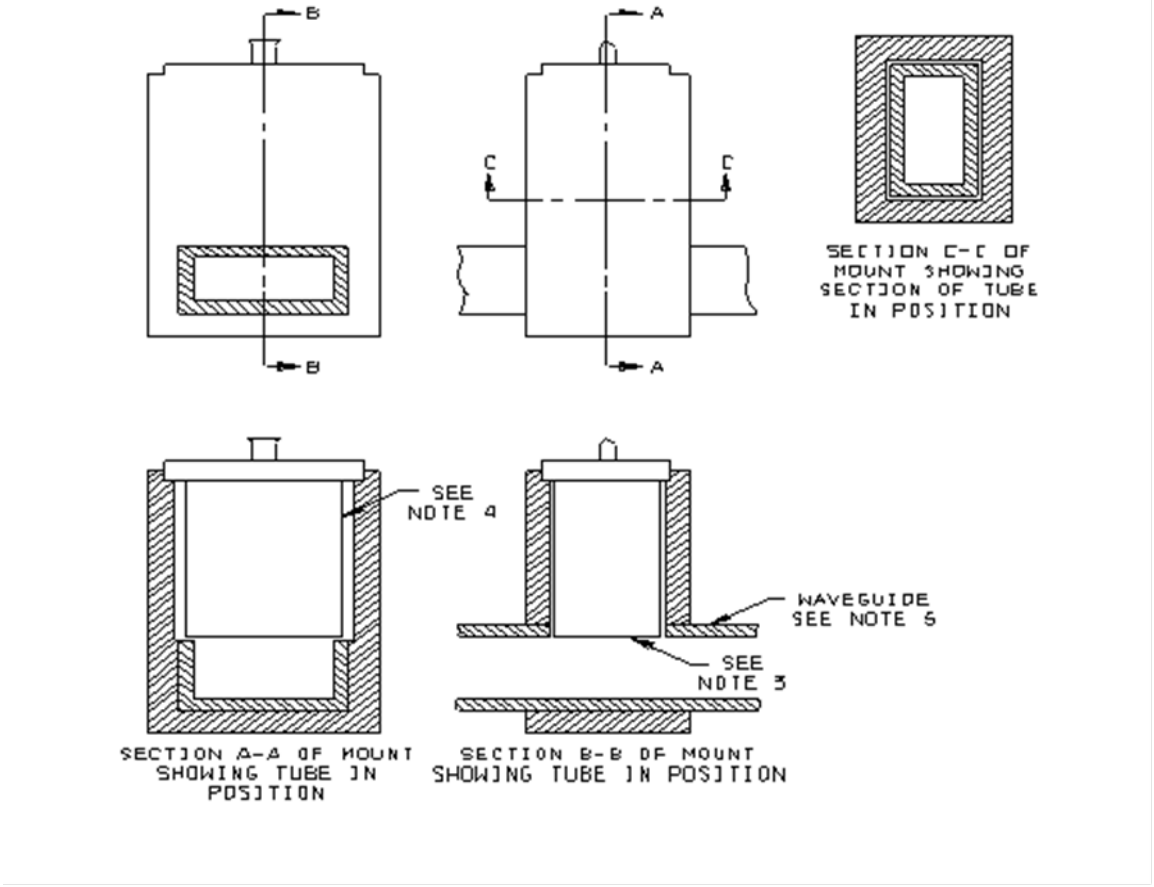
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NOTES:

1. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Tubulation shall lie within a circle 0.38 (9.65mm) diameter maximum located from centerlines of flange. Qualification inspection required.
3. Entire tube shall be plated 100 MSI minimum silver, or equivalent. Qualification inspection required.
4. Centerlines of window shall coincide with corresponding centerlines of tube body within .015 (0.38 mm) measured in plane of window. Conformance inspection, part 2, required.
5. Two gaskets as specified on Drawing 191-JAN shall be supplied unmounted. Conformance inspection, part 2 required.
6. Dimension K shall be measured 0.12 (3.05 mm) from reference plane A.
7. Dimension L shall be measured 0.12 (3.05 mm) from window plane.
8. Reference plane A shall be normal to reference plane B.
9. Flat end plate and dimpled window optional.
10. Seal-off cap over exhaust tubulation optional.
11. Position of gasket when mounted.

FIGURE 1. Outline drawing of electron tube types 1B35A , 1B37A, 6162 and 5864. -Continued.

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NOTES:

1. Design of holder optional.
2. Tube shall be held firmly in position by means of spring or screw clamps.
3. Bottom of the tube shall be flush with inside surface of waveguide within ± .004.
4. Clearance between tube walls and inner surface of mount shall be .030 to .040.
5. Waveguide shall be RG-52/U for the 1B35A, 1B37A, and 6162; and RG-51/U for the 5864. It is recommended that the RG-52/U waveguide be pressurized to prevent pulse power breakdown at at 250 kw. The RG-52/U and RG-51/U waveguides shall be in accordance with MIL-DTL-85/1.

FIGURE 2. Test mount.

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Reference documents. In addition to MIL-PRF-1 this specification sheet references MIL-STD-1311, MIL-DTL-85/1, and Drawing 191-JAN.

NOTE: To obtain copies of JAN drawings, please send a request via email to TubesAmps@dla.mil.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Navy - EC
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Preparing activity:
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(Project 5960-2024-016)

Review activities:
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